ChatGPT

Deep Sleep and Eye-First Diabetes App Strategy for Mustafa

Executive summary

For adults with type 1 or type 2 diabetes, deep non-REM sleep is relevant to eye health mostly through **indirect but clinically meaningful pathways**. Stage N3 is the deepest stage of non-REM sleep, is characterized by high-amplitude delta activity, and is the sleep period most strongly linked to physiologic restoration and the major nocturnal pulse of growth hormone. Direct human trials showing that *more N3 by itself* prevents diabetic retinopathy or diabetic macular edema are not available in the sources reviewed, but sleep loss does measurably harm the ocular surface and retinal biology: sleep deprivation reduces tear secretion, shortens tear break-up time, increases tear osmolarity, and experimental work shows disruption of the retinal circadian transcriptome, oxidative stress, microglial changes, and reduced retinal function. In diabetes, that matters because diabetic retinopathy is a microvascular eye disease, diabetic macular edema is a vision-threatening leakage/swelling complication, and sleep disturbance, long or short sleep duration, excessive daytime sleepiness, and high obstructive sleep apnea risk have all been linked with worse diabetic eye outcomes. [1]

The glucose side of the evidence is stronger. Across CGM-based studies, adults with type 1 diabetes who sleep in the recommended 7–9 hour range tend to show lower glycemic variability and higher time in range, while greater sleep timing variability is associated with spending less time in the target glucose range. Poor sleep quality is also associated with greater nocturnal glycemic variability and fear of hypoglycemia, and nocturnal hypoglycemia remains common enough in insulin-treated diabetes that it can itself become a sleep disruptor and a driver of alarm fatigue. In broader adult cohorts using CGM, persistent insufficient sleep and later sleep onset are associated with greater glycemic variability as well. [2]

For Mustafa's app challenge, the highest-value product concept is **not** "optimize sleep" in a generic way. It is to use **Libre overnight glucose patterns + fridge-agent meal suggestions + sleep timing/trend data** to reduce the specific behaviors that worsen both overnight glycemia and next-day recovery: late dinners, refined/high-glycemic evening meals, saturated-fat–heavy dinners, and unnecessary bedtime snacks. That matters because a high-glycemic meal can shorten sleep onset in healthy adults when eaten about 4 hours before bed, but dinner generally produces worse postprandial glycemia than breakfast regardless of glycemic index, and late-night dinner is associated with poorer glycemic control in type 2 diabetes. In other words, the app should optimize **sleep-supportive meals that still protect overnight glucose**, rather than importing sleep-nutrition advice developed for people without diabetes. [3]

The recommendation table below is a product synthesis from the evidence cited in this report.

Recommendation	Why it should be in the MVP	Practical implication
Favor earlier dinners	Strongest overlap between sleep and glucose evidence	Target dinner completion roughly 3–4 hours before bed when feasible
Use lower-glycemic, higher-fiber, protein-containing evening meals	Better fit for both overnight glucose stability and more restorative sleep	Fridge agent should bias toward legumes, yogurt, nuts, tofu, fish, oats, vegetables, and olive-oil-based plates
Make bedtime snacks conditional, not routine	Routine late snacks can worsen overnight glycemia; targeted snacks may reduce severe lows in insulin users	Trigger only for clinician-approved low-normal pre-bed glucose plus downward trend or recent exercise
Build overnight “phenotypes” from Libre data	Users do not all need the same meal advice	Separate late-dinner hyperglycemia, unstable overnight CV, and low-glucose-prone users
Treat eye health as an adherence and education workflow	Eye benefits are mostly indirect today	Remind about annual dilated eye exams, vision-change red flags, and sleep-disordered breathing risk
Keep the app in a support, not dosing, role initially	Best alignment with Abbott/FDA boundaries	No autonomous insulin dosing; use clinician review and user education

Evidence basis for these recommendations includes randomized meal-timing and glycemic-index studies, CGM sleep association studies, a bedtime snack trial, the 2025 DailyDose smart snack study, and Abbott/FDA documentation on how LibreView is intended to be used. [4]

Deep sleep physiology and what it means for the eye

N3 is the deepest stage of non-REM sleep, carries the highest arousal threshold, and is classically associated with physical restoration, tissue repair, immune support, and the first major sleep-related growth hormone pulse. The strongest physiology here is systemic rather than eye-specific: N3 is part of the body’s broader restorative program, but the evidence base does **not** show a simple one-step pathway such as “more N3 directly repairs the diabetic retina.” That distinction matters for app design. The app should frame N3 as a *proxy for restorative sleep* and an input into metabolic recovery, not as a stand-alone ocular therapy. [5]

The most direct eye-sleep evidence is at the ocular surface. In a controlled human study, sleep deprivation increased tear osmolarity, shortened tear break-up time, and reduced tear secretion by morning, changes that can promote ocular surface disease. A broader review of dry eye and sleep disorders similarly supports a bidirectional relationship between poor sleep and ocular surface dysfunction. For a diabetes app, this is highly relevant because diabetes already increases dry-eye risk through neuropathy, inflammation, and tear-film instability; sleep loss can add another hit. [6]

Overnight ocular hemodynamics are also dynamic. The literature reviewed shows that intraocular pressure and ocular perfusion pressure change across the 24-hour cycle and are influenced by posture and sleep state. One study using continuous contact-lens sensing reported that sleep microstructure and

macrostructure were closely related to nocturnal IOP variation, with IOP highest during REM and lower in deeper non-REM stages; older awakening-based measurements and review articles show that the exact stage pattern can differ depending on method and posture, but they consistently support the broader conclusion that nocturnal ocular pressure/perfusion is not static. That point is especially important in diabetes because abnormal ocular perfusion and abnormal vascular autoregulation are already implicated in diabetic retinopathy pathophysiology. [7]

Retinal maintenance during the sleep-wake cycle is also partly **circadian**, not just "deep-sleep-stage" dependent. Recent work shows that chronic sleep deprivation disrupts the retinal circadian transcriptome and impairs retinal function in mice, while retinal pigment epithelium phagocytic activity has a circadian peak controlled by the retinal/RPE clock. That means the app should pay attention not only to *quantity* of sleep, but to *regularity* and *timing*. A user who sleeps enough hours but does so at erratic times may still be undermining retinal and metabolic rhythms. [8]

For diabetic eye complications specifically, the relevant complication set is well defined: diabetic retinopathy is the major ocular microvascular complication of diabetes, while diabetic macular edema is a vision-threatening retinal swelling/leakage problem that may occur with retinopathy. In population data, both short and long sleep duration were associated with moderate diabetic retinopathy, and long sleep duration, excessive daytime sleepiness, and high OSA risk were associated with vision-threatening diabetic retinopathy. A meta-analysis likewise found sleep quality and duration to be linked with diabetic retinopathy risk, and a prospective cohort linked obstructive sleep apnea with diabetic macular edema and retinopathy outcomes over follow-up. Because these are associations rather than proof that N3 alteration alone changes disease course, the eye-health module should initially emphasize **screening adherence and risk education**, including annual dilated eye exams and faster review for blurred vision or visual change. [9]

Sleep quality, deep sleep, and Libre metrics

For product logic, the most useful glucose metrics are the CGM consensus metrics and the Libre reporting structure that already operationalize them. The international CGM consensus defines time in range as the percentage of readings between 70 and 180 mg/dL, with complementary time-below-range and time-above-range measures. Abbott's LibreView AGP report explicitly includes glucose statistics, targets, time in range, AGP, daily glucose profiles, low events, carbohydrate data, and insulin data. That means Mustafa's app can be built around clinically recognizable outputs instead of inventing custom-only metrics. [10]

In adults with type 1 diabetes, the strongest recent CGM sleep signal is that sleeping in the recommended 7–9 hour range is associated with lower glycemic variability and higher time in range compared with sleeping less than 7 hours or more than 9 hours. In the 2026 Frontiers study of 155 adults with type 1 diabetes using CGM, the 7–9 hour group had lower mean glucose, lower coefficient of variation, lower MAGE and MODD, and higher TIR; their mean CV was the only one within the ≤36% consensus target. This is a design-friendly result because it suggests there is a pragmatic sweet spot for sleep duration that can be communicated simply in-app. [11]

Sleep regularity appears at least as important as sleep duration. In adults with type 1 diabetes, greater sleep timing variability was independently associated with less time spent in the desirable glucose range, and the study highlighted a meaningful link between variable sleep and worse CGM-derived glucose control. In practical terms, this means the app should reward **consistent sleep timing**, not only total sleep

minutes. A refrigerator agent that optimizes food while ignoring bedtime drift or social jetlag would miss a major lever. [12]

Sleep and overnight glucose also interact in the other direction. In adults with type 1 diabetes, poor sleep quality was associated with greater nocturnal glycemic variability and with fear of hypoglycemia in a pooled analysis using 3-day CGM. Reviews and pilot work in the CGM era further show that nocturnal hypoglycemia remains common, can be prolonged, and may degrade subjective sleep quality and next-day cognitive performance. This is exactly the use case where a combined Libre + sleep + meal feature can outperform a generic meal-planning app: it can distinguish users whose sleep is being disrupted by biochemical instability from those whose sleep issue is primarily timing or sleep hygiene. [13]

The same pattern is now appearing in broader adult cohorts and in type 2 diabetes, although the literature is newer. In a 2025 JAMA Network Open cohort, persistent inadequate sleep duration and late sleep onset were associated with greater CGM-derived glycemic variability in middle-aged and older adults. Separately, recent work in adults with type 2 diabetes found that poor self-reported sleep quality was associated with less favorable CGM-derived variability and range metrics in a 14-day blinded-CGM design. For Mustafa's app, that means the concept is not narrowly limited to type 1 diabetes, but the **strongest overnight-risk use case remains insulin-treated adults**, especially those susceptible to nocturnal lows or wide overnight excursions. [14]

Meals that are most likely to support restorative sleep and stable overnight glucose

The meal-timing signal is strong enough to justify a dedicated product feature. Randomized work shows that carbohydrate-rich meals eaten at dinner produce greater postprandial glycemia than similar meals eaten at breakfast, regardless of glycemic index. Observational and interventional studies further suggest that late-night dinner is associated with poorer glycemic control in type 2 diabetes, while shifting dinner earlier can improve fasting and overall glycemic measures in habitual late eaters. For app logic, "when" dinner happens should be treated as a first-class variable, not as a note attached to nutrient totals. [15]

Macronutrient composition matters for both sleep and glucose. Sleep research shows that lower fiber and higher saturated fat and sugar intake are associated with lighter, less restorative sleep and more arousals, with less slow-wave sleep in some analyses. Reviews of diet and sleep broadly converge on the same direction: healthier dietary patterns, especially Mediterranean-style patterns rich in vegetables, legumes, fruits, whole grains, nuts, olive oil, and fish, are associated with better sleep quality, while processed and free-sugar-rich diets are associated with worse sleep. Diabetes nutrition reviews also support Mediterranean and low-GI/low-GL eating patterns as effective options for glycemic management. The overlap between these literatures is the app's main nutritional opportunity. [16]

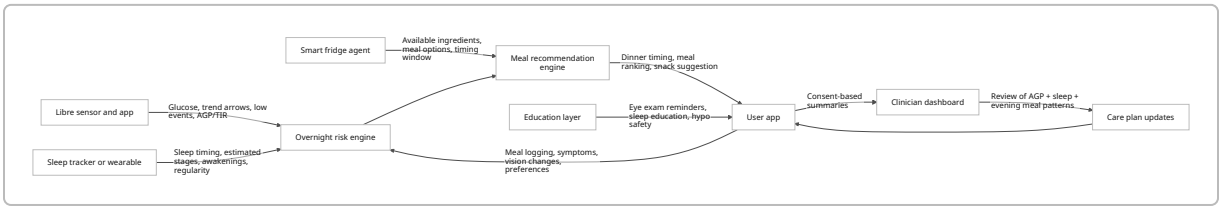
One nuance is important: a high-glycemic meal eaten about 4 hours before bed can shorten sleep onset in healthy adults, but it is not a good default strategy for people with diabetes. The same literature that makes high-GI sleep advice sound attractive is offset by diabetes-relevant data showing that dinner glycemia is worse than breakfast glycemia and that high-GI meals worsen postprandial glucose irrespective of timing. For Mustafa's fridge agent, the correct rule is not "high GI helps sleep," but rather "if a meal is close to bedtime, bias toward *moderate carbohydrate quality plus fiber/protein/unsaturated fat* and earlier timing." [17]

Bedtime snacks should therefore be **conditional**. Older randomized work in type 1 diabetes found that the need for, and ideal composition of, a bedtime snack depended on bedtime glucose level; standard or protein-containing snacks could prevent nocturnal hypoglycemia at lower bedtime glucose levels. Newer digital-intervention work is even more relevant: the 2025 DailyDose Smart Snack crossover study used CGM and activity inputs to personalize bedtime snack recommendations and did not reduce all <70 mg/dL overnight lows, but it did reduce nocturnal events <54 mg/dL without worsening other glycemic metrics. At the same time, routine bedtime snacking can worsen nocturnal glycemia in some groups, especially when used reflexively instead of selectively. That argues for a rules engine that recommends pre-bed snacks only when the glucose trajectory and risk context justify them. [18]

Specific “sleep foods” should be used carefully. Reviews suggest potential sleep benefits from kiwifruit, tart cherry, and other foods containing melatonin-, serotonin-, or polyphenol-related compounds, but most of that evidence comes from non-diabetic or mixed populations. For a diabetes app, these foods should be treated as **optional, CGM-conditional ingredients**, not default bedtime prescriptions. A small serving dose that produces no clinically important overnight rise for one user may be unacceptable for another. That is where Libre-linked personalization matters more than generic nutrition folklore. [19]

Product blueprint for Mustafa’s app

The right product framing is: **“Use overnight glucose patterns to personalize evening eating in ways that improve restorative sleep and support eye health.”** Abbott’s current ecosystem already supports historical glucose review, time-in-range reporting, automatic uploads from supported apps, remote sharing with clinicians, and secure cloud storage through LibreView, but Abbott also states that LibreView is a secondary viewer and is not intended for immediate clinical decision-making or to replace professional advice. That is actually helpful for product scope: Mustafa’s app should operate as an intelligent behavioral layer on top of Libre data, not as an autonomous treatment engine. [20]



The highest-priority features are the ones with the tightest evidence loop between evening behavior and overnight CGM outcomes. In practice, that means: first, an **overnight phenotype engine** that classifies users into patterns such as “late-dinner hyperglycemia,” “downward-trending pre-bed glucose,” “high overnight variability,” or “frequent low alarms”; second, a **dinner timing coach** that nudges earlier dinner completion; third, a **meal builder** that scores fridge ingredients for both glycemic stability and sleep support; and fourth, a **conditional bedtime snack advisor** for clinically appropriate users. Those four features directly target the pathways most consistently supported by current evidence. [21]

Priority	Feature	Rationale	Product rule
Highest	Overnight phenotype engine	Different CGM patterns need different meal advice	Build nightly classification from TIR, TBR, TAR, trend at bedtime, overnight CV, and low-event history
Highest	Dinner timing coach	Earlier dinner is the most scalable metabolic lever	Nudge dinner earlier when repeated overnight highs follow late meals
Highest	Fridge-agent dual score	Ingredient recommendations should reflect both sleep and glucose	Rank meals using fiber/protein/unsaturated-fat positives and late refined-carb/saturated-fat penalties
Highest	Conditional bedtime snack advisor	Some users need protection from severe lows; many do not need routine snacks	Offer snack only when risk thresholds are met and user has clinician-approved settings
Medium	Smart alert orchestration	Reduce alarm fatigue and protect sleep	Suppress low-value nudges at night; escalate only for meaningful predicted risk
Medium	Eye-health workflow	Eye benefits are largely indirect today but clinically important	Remind for annual dilated exams, vision-change red flags, and OSA risk conversations
Medium	Weekly clinician summary	Makes the app credible in real care	Export a combined AGP + sleep regularity + dinner-time report
Lower	Reinforcement learning from response	Improves personalization after launch	Re-rank meals by observed overnight response, not only nutrient estimates

The educational layer should be concise and condition-specific. A user whose recurring problem is overnight downward drift needs education about recent exercise, alcohol, insulin timing, and selective snacks; a user whose recurring problem is late-evening hyperglycemia needs education about earlier dinner, carbohydrate quality, and portion size; a user with worsening daytime sleepiness, snoring, and unstable glucose should receive a prompt to discuss sleep-disordered breathing with a clinician because OSA has implications for both glycemia and diabetic eye disease. The app should also include annual dilated eye exam reminders because eye protection in this product is most defensible today as **risk reduction plus earlier screening adherence** rather than as direct disease treatment. [22]

Implementation, privacy, regulation, integration, and validation

The privacy model should be built around **explicit user authorization, minimum necessary data use, and separated consent flows** for consumer coaching versus clinician sharing. Abbott's Libre privacy documents state that Libre products collect and process personal information to provide Libre services, and LibreView is described as a secure, cloud-based system; Abbott also documents two-factor authentication for care-

team access. If Mustafa's app interfaces with covered entities or business associates, HIPAA obligations may apply for ePHI, but HHS is also clear that HIPAA does not automatically protect all health information stored on personal devices outside covered-entity/business-associate relationships, and HHS has warned covered entities about online tracking technologies in apps and websites. Practically, that means the app should minimize third-party SDKs, avoid ad-tech trackers in regulated workflows, and separate consumer analytics from identifiable clinical data. [23]

The regulatory line is manageable if the product is disciplined. FDA's 2026 General Wellness guidance supports low-risk products that promote healthy lifestyle behaviors, and the 2026 Clinical Decision Support guidance clarifies when CDS functions may be excluded from device regulation and when software functions remain devices. Mustafa's app is most likely to stay on the safer side of that line if it emphasizes education, pattern visualization, consensual summaries, and behavioral suggestions such as meal timing and ingredient selection. The regulatory risk rises if the app starts making patient-specific treatment recommendations intended to diagnose, prevent, or treat diabetic retinopathy, macular edema, hypoglycemia, or insulin dosing without transparent clinician interpretability and appropriate controls. A prudent launch strategy is therefore to position the eye-health pathway as **education + screening adherence + symptom prompting**, not as retinal diagnosis. [24]

Interoperability should respect both what Libre officially supports and what the app actually needs. Abbott's current public materials show minute-by-minute automatic glucose updates in supported Libre apps, automatic syncing to LibreView when connectivity is available, remote sharing with clinicians via Practice IDs/invitations, AGP/TIR reporting, and different upload behavior for app users versus reader users. LibreView is historical and secondary, not an immediate-treatment interface. From an engineering standpoint, the app should therefore be comfortable with **near-real-time consumer data for coaching** and **historical summaries for clinician workflows**. If EHR integration becomes important, the emergence of an HL7 FHIR continuous glucose monitoring implementation guide strengthens the case for standards-based downstream interoperability rather than custom PDF exchange. [25]

The UX should be designed for diabetic users who already manage cognitive load and nighttime vigilance. CGM low alarms are valuable, but excessive alerts can fragment sleep and amplify fear of hypoglycemia. The app should therefore use tiered notifications: educational meal nudges in the evening, quiet overnight monitoring by default, and escalation only for high-risk predicted lows or repeated clinically important patterns. Users should also be able to choose whether the sleep layer focuses on *regularity*, *overnight lows*, or *late-dinner highs*, because the same dashboard will otherwise feel noisy and irrelevant. This is especially important for insulin-treated users, where poor sleep quality, nocturnal glycemic variability, and fear of hypoglycemia can reinforce one another. [26]

A sensible clinical validation program would have three phases. First, a **feasibility study** should test onboarding, sensor syncing, meal logging burden, and user adherence to earlier dinner nudges. Second, a **micro-randomized trial** can test whether just-in-time meal-timing prompts or snack recommendations improve next-night CGM outcomes. Third, a **parallel-group randomized trial** can compare standard Libre use versus Libre + fridge-agent + sleep optimization. Core glycemic endpoints should include TIR, overnight TBR <70 mg/dL and <54 mg/dL, TAR, mean overnight glucose, and CV, all aligned with consensus CGM metrics and LibreView reporting. Sleep outcomes should include regularity, total sleep time, and—if the team wants to make a bold "deep sleep" claim—estimated N3 from a validated wearable plus a polysomnography substudy in a subset, because consumer devices alone are better suited to trend tracking than to definitive physiologic claims. Eye endpoints should be modest and realistic for an app challenge:

annual dilated eye-exam completion, symptom-prompt adherence, dry-eye symptom scores, and—only in longer studies—retinal imaging or ophthalmology referral outcomes. [27]

Sample week and ingredient comparison

The sample plan below assumes an adult with type 1 or type 2 diabetes, access to common refrigerator ingredients, and a goal of **earlier dinners, steadier overnight glucose, and more restorative sleep**. It favors Mediterranean/low-GI principles, high fiber, moderate lean protein, unsaturated fat, and limited refined starch late in the day. It also assumes that bedtime snacks are **optional and conditional**, especially for insulin-treated users. [28]

Day	Breakfast	Lunch	Dinner	Optional pre-bed snack	CGM pattern this day is designed for
Monday	Greek yogurt, chia, berries, walnuts	Lentil salad with cucumber, tomato, olive oil, feta	Baked salmon, broccoli, quinoa	None unless low-normal pre-bed glucose with downward trend	Baseline steady day
Tuesday	Oats with cinnamon, flax, unsweetened milk	Turkey or tofu wrap with hummus and greens	Chicken or tofu stir-fry with non-starchy vegetables and brown rice, eaten early	Small protein-forward snack only if indicated	Late-dinner hyperglycemia prevention
Wednesday	Eggs with spinach and avocado	Bean-and-vegetable soup with side salad	Chickpea pasta or lentil bowl with olive oil vegetables	None	High overnight variability reduction
Thursday	Cottage cheese or unsweetened yogurt with pumpkin seeds	Tuna or edamame grain bowl	Cod or tofu, cauliflower mash, green beans	Small yogurt or nuts if post-exercise and trending down	Post-exercise low-risk night

Day	Breakfast	Lunch	Dinner	Optional pre-bed snack	CGM pattern this day is designed for
Friday	Overnight oats with chia and peanut butter	Leftover salmon/tofu salad	Turkey meatballs or tempeh with roasted vegetables and modest whole grain	None	End-of-week routine stabilization
Saturday	Veggie omelet with side berries	Chicken/bean salad with olive oil dressing	Homemade chili with beans, extra vegetables, small whole-grain side	Protein-containing snack only if clinician-approved rule triggers	Prevent weekend late-night snacking
Sunday	Yogurt bowl with nuts and kiwi if prior response was acceptable	Mediterranean leftovers plate	Earlier lighter dinner: soup + salad + grilled fish/tofu + small carb portion	None in most users	Reset sleep timing before work week

How the app should adapt this plan is more important than the plan itself. If Libre shows repeated overnight highs after rice, the fridge agent should down-rank rice portions at dinner and shift carbohydrate earlier in the day. If Libre shows pre-bed downward trends after evening exercise, the app should consider a clinician-approved protein-containing snack pathway. If the user sleeps poorly despite flat glucose, the next intervention should target dinner timing, caffeine/alcohol, and schedule regularity rather than adding more bedtime carbohydrate. Those adjustments are the essence of the product. [29]

The ingredient table below is a synthesis for fridge-agent ranking. “Sleep impact” means likely fit with restorative sleep based on the cited dietary literature; “glycemic impact” means likely fit for overnight glucose stability in typical adults with diabetes, not a guarantee for every user.

Ingredient or food pattern	Likely sleep impact	Likely overnight glycemic impact	Best app use
Greek yogurt, cottage cheese, unsweetened kefir	Favorable	Usually favorable	Base for protein-forward evening snack if needed
Nuts and seeds	Favorable	Favorable in modest portions	Add to yogurt, salads, oats

Ingredient or food pattern	Likely sleep impact	Likely overnight glycemic impact	Best app use
Legumes, lentils, beans, chickpeas	Favorable	Favorable	Strong dinner anchor; high fiber and lower glycemic load
Non-starchy vegetables	Favorable	Favorable	Increase dinner volume without late glucose surge
Fish, tofu, eggs, lean poultry	Favorable to neutral	Favorable	Main protein options for stable evenings
Oats, quinoa, modest intact whole grains	Neutral to favorable	Generally favorable if portioned	Better dinner carb than refined starch
Olive oil, avocado	Favorable in moderation	Favorable	Preferred fat sources
Kiwifruit or tart-cherry products	Potential sleep benefit, but evidence mostly outside diabetes	Mixed; individualized	Optional only after CGM-confirmed tolerance
White rice, refined bread, sweet desserts at dinner	May shorten sleep onset in some settings but not ideal overall	Often unfavorable	Down-rank for evening use
Fried, saturated-fat–heavy, very large dinners	Unfavorable	Often unfavorable	Strong negative weight in meal ranking
Alcohol close to bedtime	Unfavorable	Can destabilize overnight glucose, especially in insulin users	Red-flag or discourage near bedtime

This ranking is grounded in the overlap between diet-sleep and diet-glycemia evidence: fiber- and unsaturated-fat–rich patterns are associated with better sleep quality and less fragmented sleep; Mediterranean-style patterns fit both sleep and diabetes goals; low-GI/low-GL approaches support glycemic management; and targeted, not routine, bedtime snacks are the safer rule for insulin-treated users. The important caveat is that “sleep foods” with intrinsic sugars should only be promoted when the user’s own Libre response makes them acceptable. [30]

The bottom line for Mustafa’s app is that the winning concept is not a generic sleep tracker and not a generic meal planner. It is a **closed behavioral loop**: use Libre to infer the user’s overnight glucose phenotype, use the fridge agent to generate earlier and more sleep-compatible evening meals from what is actually available, and use the sleep layer to keep the system focused on regularity and restoration. That strategy is evidence-based, product-differentiated, compatible with Abbott’s current historical-data framing, and credible for an app challenge centered on diabetes self-management. [31]

1 5 Physiology, Sleep Stages - StatPearls - NCBI Bookshelf
https://www.ncbi.nlm.nih.gov/books/NBK526132/

2 11 21 Frontiers | Adults with type 1 diabetes who sleep 7–9 hours per night present lower glycemic variability: a cross-sectional study
https://www.frontiersin.org/journals/endocrinology/articles/10.3389/fendo.2025.1745272/full

3 17 High-glycemic-index carbohydrate meals shorten sleep onset - PubMed
https://pubmed.ncbi.nlm.nih.gov/17284739/

4 15 28 29 High or low glycemic index (GI) meals at dinner results in greater postprandial glycemia compared with breakfast: a randomized controlled trial - PubMed
https://pubmed.ncbi.nlm.nih.gov/32327444/

6 Sleep deprivation reduces tear secretion and impairs the tear film - PubMed
https://pubmed.ncbi.nlm.nih.gov/24833736/

7 Relationship Between Nocturnal Intraocular Pressure ...
https://pubmed.ncbi.nlm.nih.gov/26505463/?utm_source=chatgpt.com

8 Chronic sleep deprivation impairs retinal circadian transcriptome and visual function - PubMed
https://pubmed.ncbi.nlm.nih.gov/38649019/

9 Diabetic Retinopathy - StatPearls - NCBI Bookshelf - NIH
https://www.ncbi.nlm.nih.gov/books/NBK560805/?utm_source=chatgpt.com

10 27 Clinical Targets for Continuous Glucose Monitoring Data ...
https://diabetesjournals.org/care/article/42/8/1593/36184/Clinical-Targets-for-Continuous-Glucose-Monitoring?utm_source=chatgpt.com

12 Association between sleep variability and time in range of ...
https://pubmed.ncbi.nlm.nih.gov/37709596/?utm_source=chatgpt.com

13 26 Poor sleep quality is associated with nocturnal glycaemic variability and fear of hypoglycaemia in adults with type 1 diabetes - PubMed
https://pubmed.ncbi.nlm.nih.gov/29917259/

14 Trajectories of Sleep Duration, Sleep Onset Timing, and ...
https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2831009?utm_source=chatgpt.com

16 30 Fiber and Saturated Fat Are Associated with Sleep Arousals ...
https://pmc.ncbi.nlm.nih.gov/articles/PMC4702189/?utm_source=chatgpt.com

18 Impact of bedtime snack composition on prevention ...
https://pubmed.ncbi.nlm.nih.gov/12502652/?utm_source=chatgpt.com

19 Nutritional Interventions for Enhancing Sleep Quality - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC12678061/?utm_source=chatgpt.com

20 31 LibreView Quick Start Guide for Patients
https://pat.libreview.io/articles/qsg/

22 Nocturnal Hypoglycemia in the Era of Continuous Glucose ...
https://pmc.ncbi.nlm.nih.gov/articles/PMC11418455/?utm_source=chatgpt.com

23 PRIVACY NOTICE
https://api.libreview.io/document/pnllu?lang=en

24 General Wellness: Policy for Low Risk Devices - Guidance
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/general-wellness-policy-low-risk-devices?utm_source=chatgpt.com

25 Diabetes Products and Solutions | Manage Diabetes with CGM| Abbott U.S.
https://www.abbott.com/en-us/products-solutions/diabetes